

UART PROJECT

TEAM NAME : **BJT**

TEAM MEMBERS:

- Vundyala Bhavitha
- Boddu Joshnavi
- Amanaganti Tejasri

Objective

To design and implement a UART system using Verilog that includes a baud rate generator, framing logic, and parity-based error detection, and to verify its operation using various test cases to ensure proper timing between transmitter and receiver.

Design Configuration

Parameter	Value
HDL	Verilog HDL
System clock	50 MHz
Baud rate	9600 baud
Receiver sampling	16× oversampling
Data	8 bits, LSB first
Parity	Odd Parity
Stop bit	1

1.UART Frame Format

startbit (logic 0)	D0	D1	D2	D3	D4	D5	D6	D7	parity bit (optional)	stop bit 1 (logic 1)
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- Start bit: Logic 0; indicates the beginning of a frame.
- Data bits: 8 bits; D0 is transmitted first.
- Parity bit: Odd parity generated from the eight data bits.
- Stop bit: Logic 1; indicates the end of the frame

Frame length = 1 start + 8 data + 1 parity + 1 stop = 11 bits

2. Baud-Rate Generation

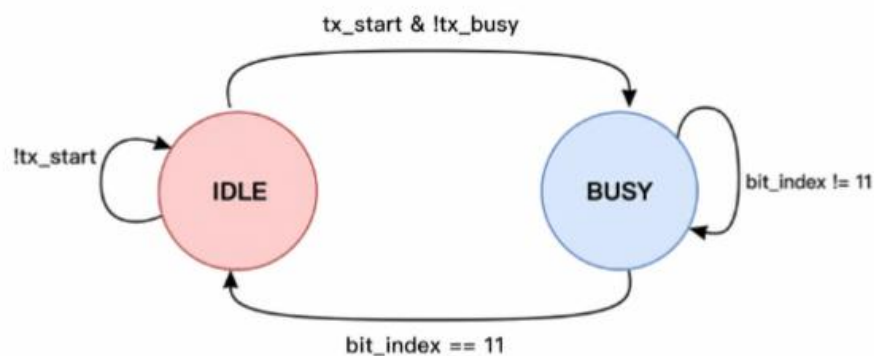
Baud Rate = The number of bits transmitted/received per second.

Baud-rate generator derives two timing enables from the 50 MHz system clock: baud tick for the transmitter and $16\times$ sampling tick for the receiver.

It is calculated by:

- **Clk_cycles_tx** $\rightarrow$ Number of clock cycles for transmitting a bit
- **Clk_cycles_tx** = **System clock frequency** / **Baud rate**
- **Clk_cycles_rx** $\rightarrow$ Number of clock cycles per oversampling tick
- **Clk_cycles_rx** = **System clock frequency** / (**$16 \times$ Baud rate**)

3 .UART Transmitter



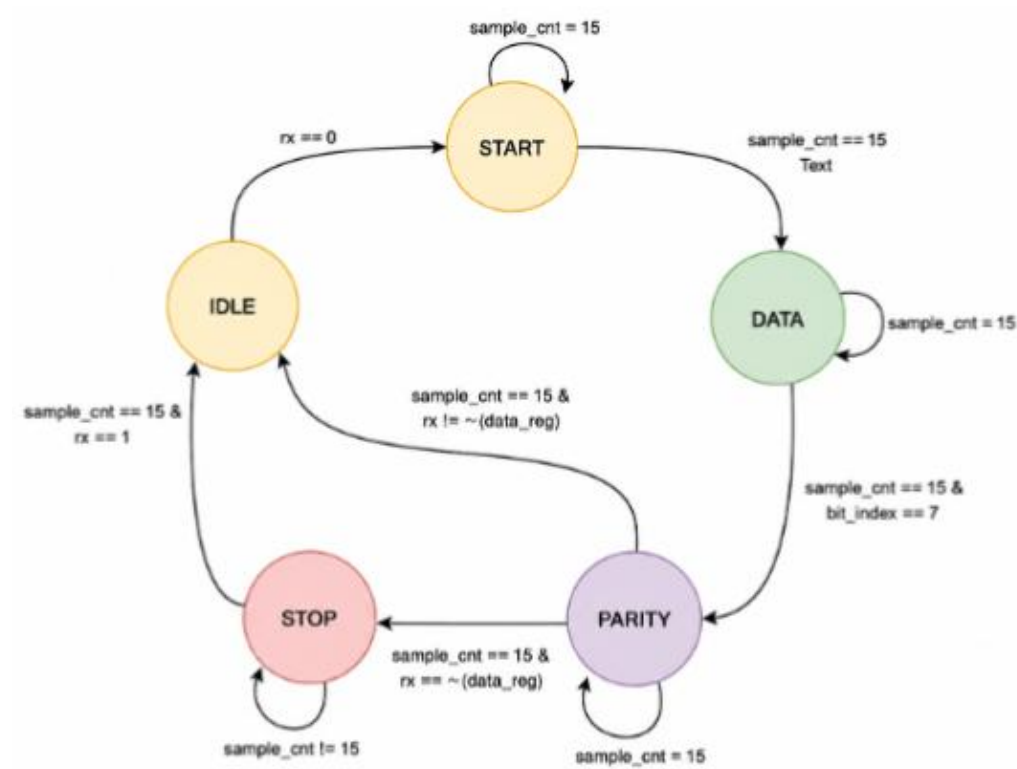
tx_start $\rightarrow$ starts transmission

tx_data $\rightarrow$ Input parallel data

tx_busy $\rightarrow$ Indicates transmission is in progress

tx $\rightarrow$ Serial output line from transmitter

3.UART Receiver

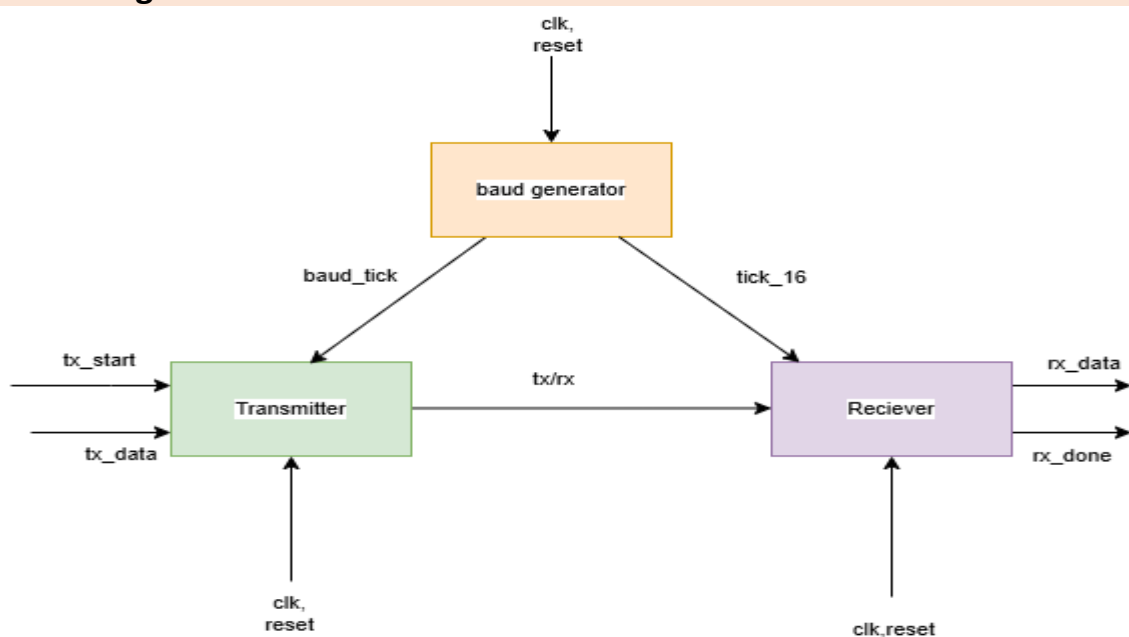


`rx` → Serial input line of receiver

`rx_data` → Received data

`rx_done` → Indicates a frame of data is received

Block diagram of UART



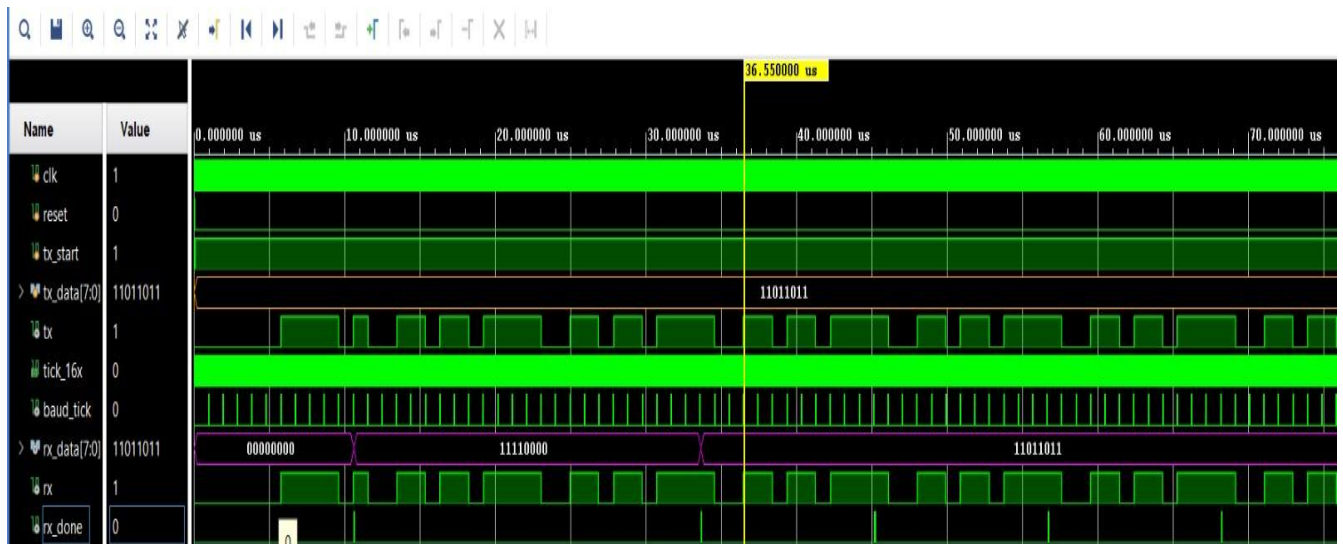
TEST CASES

1.2-Byte continuos transmission

Applied signals

tx_data=8'b11110000

tx_data=8'b11011011(2nd data byte)



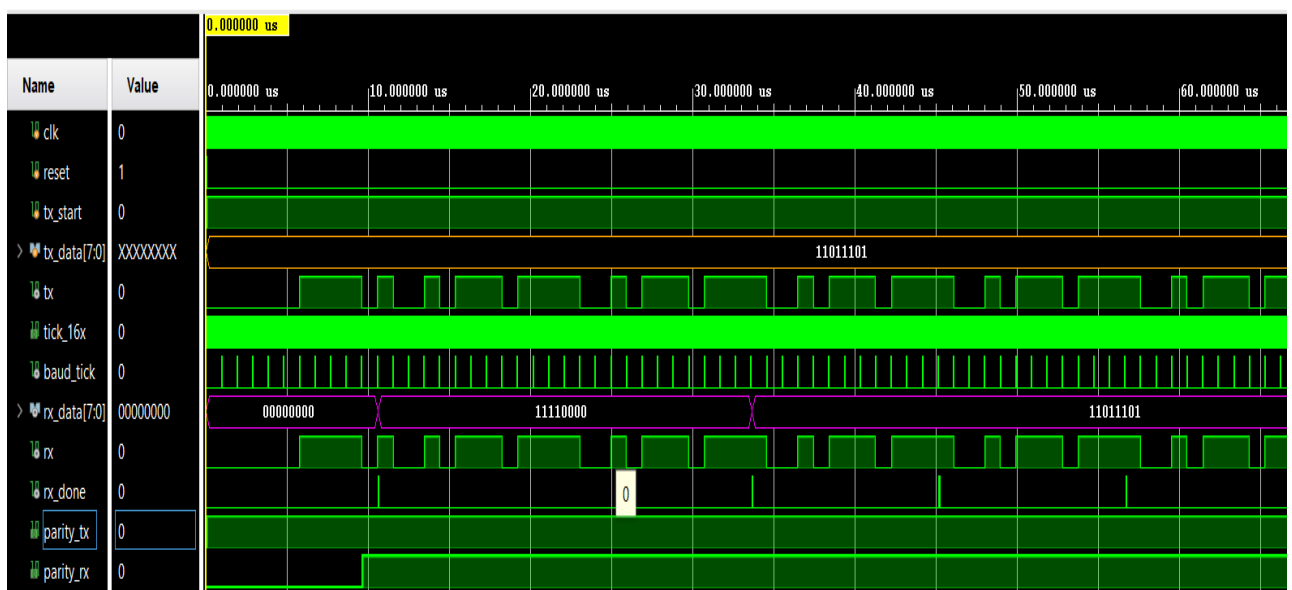
2.Error detection (parity based)

odd parity= $\sim(^{\text{data_reg}})$

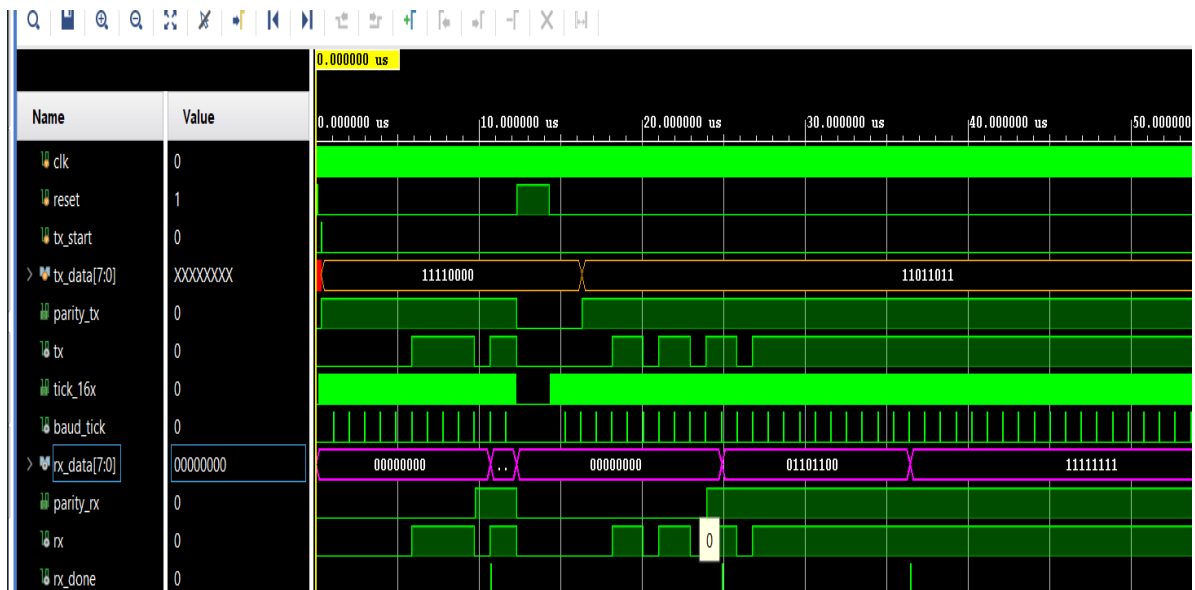
1st data byte is 11110000

Parity=1

2nd data byte is 11011011 Parity=1



3.Reset applied between 2 frames



Because reset is asserted during transmission, the receiver is reset and loses frame synchronization, resulting in incorrect data reception.

Conclusion

The UART communication system was successfully designed and implemented using Verilog HDL. The design includes a baud-rate generator, UART transmitter and 16× oversampling receiver. The system was verified using continuous two-byte transmission, parity-based error detection and reset testing.